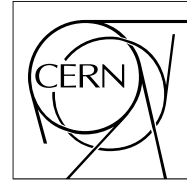


**The Compact Muon Solenoid Experiment**

# **CMS Performance Note**

Mailing address: CMS CERN, CH-1211 GENEVA 23, Switzerland



**19 October 2025**

## **Comparison of the performance of tau reconstruction and identification algorithms in Run 3**

CMS Collaboration

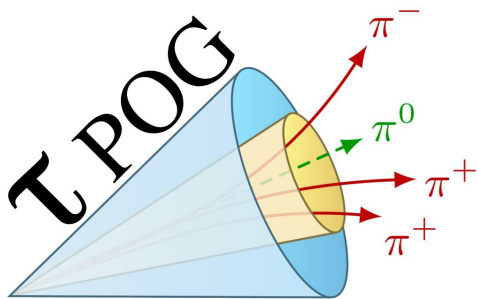
### **Abstract**

In recent years, the CMS experiment has made several advancements in the identification of hadronically decaying tau leptons ( $\tau_h$ ) using advanced machine learning (ML) techniques. This note provides a comparison between the three ML-based  $\tau_h$  identification algorithms deployed by the CMS Collaboration during the LHC Run 3: DeepTau, PNet, and UParT. The combined strength of these algorithms allows to achieve the best identification performance of  $\tau_h$  to date.

# Comparison of the performance of tau reconstruction and identification algorithms in Run 3

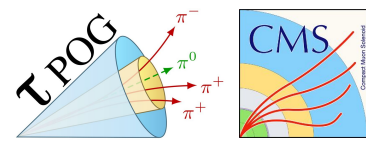
CMS Collaboration

Contact: [cms-phys-conveners-TAU@cern.ch](mailto:cms-phys-conveners-TAU@cern.ch)



# Abstract

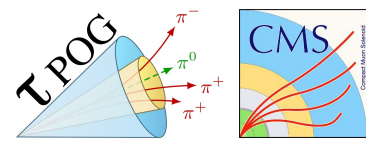
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In recent years, the CMS experiment has made several advancements in the identification of hadronically decaying tau leptons ( $\tau_h$ ) using advanced machine learning (ML) techniques. This note provides a comparison between the three ML-based  $\tau_h$  identification algorithms deployed by the CMS Collaboration during the LHC Run 3: DeepTau, PNet, and UParT. The combined strength of these algorithms allows to achieve the best identification performance of  $\tau_h$  to date.

# Glossary (1/3)

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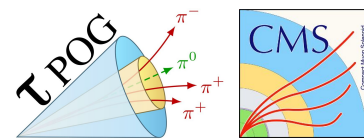


**AK4 jets:** Jets that are reconstructed by the anti-kt algorithm [1] with a distance parameter of  $R=0.4$  using particle-flow candidates [2].

**CHS jets:** AK4 jets with pileup mitigated by subtraction of charged-hadrons which do not originate from the leading vertex (CHS) [3].

**PUPPI jets:** AK4 jets with pileup mitigated by technique called the pileup-per-particle identification (PUPPI) algorithm [3,4,5], which assigns a weight to every particle depending on its probability to originate either from the leading or a pileup vertex.

# Glossary (2/3)

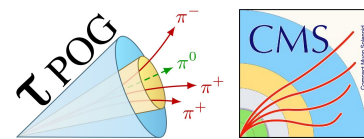


**Hadronic taus ( $\tau_h$ ):** products of tau lepton decays to hadrons excluding not detected neutrinos.

**Misidentified taus (misID  $\tau_h$ ):** reconstructed and identified hadronic tau candidates originating from quark or gluon jets ( $\text{jet} \rightarrow \tau_h$ ), electrons ( $e \rightarrow \tau_h$ ), or muons ( $\mu \rightarrow \tau_h$ ).

**HPS taus:** hadronic tau candidates reconstructed from AK4 jets constituents by the hadrons-plus-strip (HPS) algorithm [6,7], which combines charged hadrons ( $h^\pm$ ) with  $\pi^0$  candidates reconstructed as eta-phi clusters of photons and electrons (strips). The HPS algorithm reconstructs four main decay modes:  $h^\pm$ ,  $h^\pm\pi^0$ ,  $h^\pm 2\pi^0$ s,  $h^\pm h^\mp h^\pm$ ,  $h^\pm h^\mp h^\pm \pi^0$ s, and two decay modes with  $h^\pm h^\mp (\pi^0$ s) (“2-prong”) recovering  $h^\pm h^\mp h^\pm (\pi^0$ s) candidates with non-reconstructed  $h^\pm$ . Due to the stringent mass requirements for reconstructed  $\tau_h$  candidates, which are consistent with the masses of the intermediate resonances, the HPS algorithm rejects approximately 80% of misidentified quark or gluon jets, reducing its efficiency to around 75%.

# Glossary (3/3)

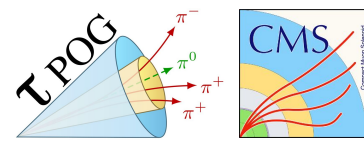


**DeepTau:** A Convolutional Neural Network architecture [7,8] customised for HPS tau candidate classification to identify genuine hadronic tau candidates and misidentified quark and gluon jets, electrons and muons. DeepTau v2.5 for Run 3 is discussed in this note [8].

**ParticleNet (PNet):** A Graph Neural Network architecture [9] customised for AK4 jets, to perform in an inclusive way jet flavour and hadronic tau identification combined with a flavour aware jet energy regression [10] and jet energy resolution estimation. Hadronic tau candidates are classified accordingly to a decay mode and charge hypothesis.

**UParT:** A Particle Transformer [11,12,13] model designed for AK4 jet, performing in an inclusive way jet flavour and hadronic tau identification combined with a flavour aware jet energy regression and jet energy resolution estimation [12]. Hadronic tau candidates are classified accordingly to a decay mode and charge hypothesis; UParT v2 is discussed in this note.

# Introduction

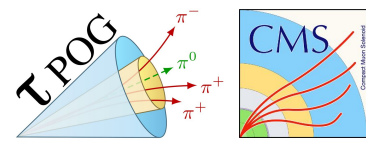


This note compares the performance of algorithms deployed by the CMS Collaboration to identify hadronically decaying tau leptons ( $\tau_h$ ) — DeepTau v2.5, PNet, UParT v2.

Since the algorithms have been originally defined on three different reconstructed objects — HPS  $\tau_h$  candidates, CHS jets, and PUPPI jets — their efficiency and purity are evaluated on a set of  $\tau_h$  candidates defined as an union of the different reconstructed objects with  $p_T > 20$  GeV,  $|\eta| < 2.5$  and  $|d_z| < 0.2$  cm. The unified collection is built starting from HPS  $\tau_h$  candidates then by adding CHS jets and then PUPPI jets not overlapping with previously added candidates. CHS and PUPPI jets are non-corrected for energy scale defined for quark and gluon jets and overestimating  $p_T$  of  $\tau_h$ . The longitudinal impact parameter ( $d_z$ ) of a  $\tau_h$  candidate is defined as  $d_z$  of its leading charged particle with respect to leading vertex.

The performance of the classifiers is then evaluated taking into account also the reconstruction efficiency of their related reconstruction algorithm.

# Definitions



## Discriminants:

- **vsJet (against jet  $\rightarrow \tau_h$  misID):**  $\sum(\tau_h \text{ scores}) / \sum(\tau_h \text{ \& all jet-flavour scores})$
- **vsE (against e  $\rightarrow \tau_h$  misID):**  $\sum(\tau_h \text{ scores}) / \sum(\tau_h \text{ \& electron scores})$
- **vsMu (against  $\mu \rightarrow \tau_h$  misID):**  $\sum(\tau_h \text{ scores}) / \sum(\tau_h \text{ \& muon scores})$

**Charge assignment confidence (qConf):**  $(\sum(\tau_h^+ \text{ scores}) - \sum(\tau_h^- \text{ scores})) / (2 * \sum(\tau_h \text{ scores}))$

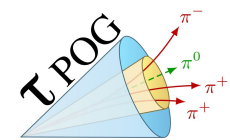
**Tight charge confidence criteria:** Exclude “2-prong” HPS candidates and no-HPS candidates with  $|qConf| < 0.2$  (using native qConf for tested algorithm)

## Efficiency and misID probability:

- **Denominator:** union of differently reconstructed  $\tau_h$  candidates with  $p_T > 20 \text{ GeV}$ ,  $|\eta| < 2.5$ ,  $|d_z| < 0.2$
- **Numerator:** denominator & vsJet/E/Mu > threshold & charge confidence criteria

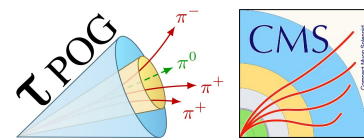
**Samples:** NLO DY( $\rightarrow e, \mu, \tau$ )+jets,  $50 < m(\ell\ell) < 120 \text{ GeV}$  with Powheg+Pythia8 for genuine  $\tau_h$  and jet, e,  $\mu \rightarrow \tau_h$  misID



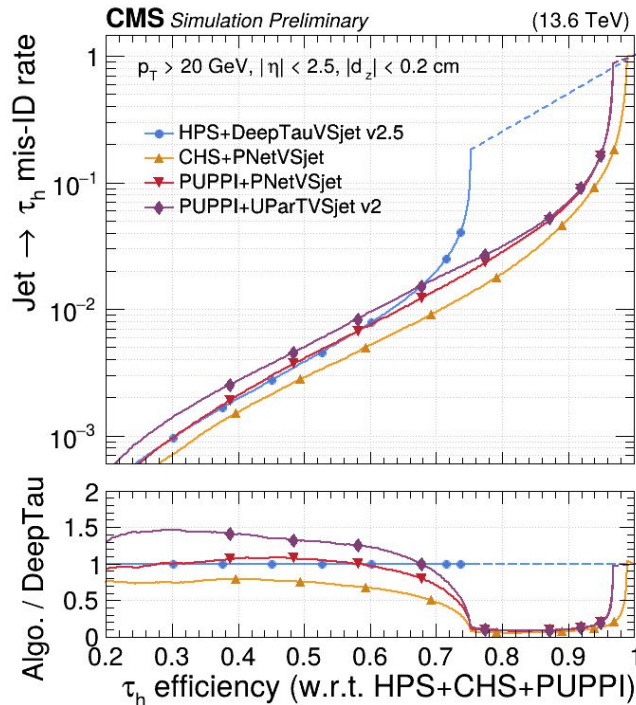


# Performance of hadronic taus identification

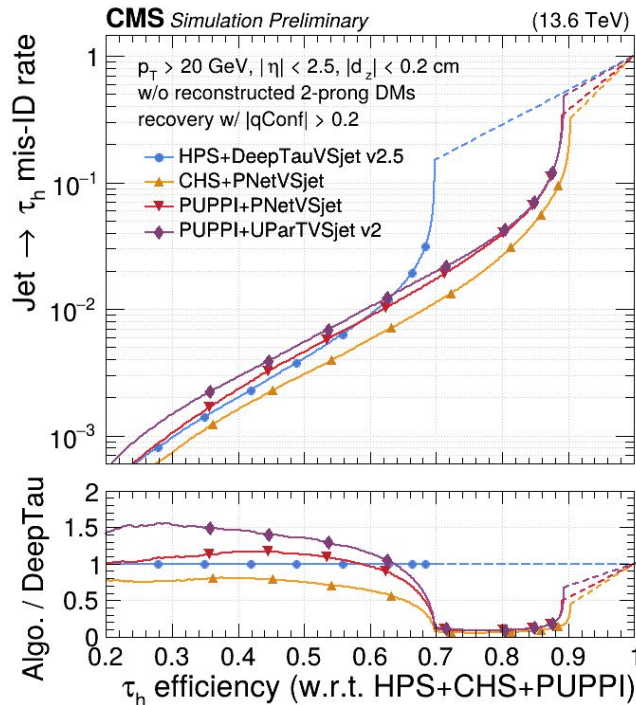
# Discrimination vs. jet $\rightarrow \tau_h$ misID



## No charge confidence



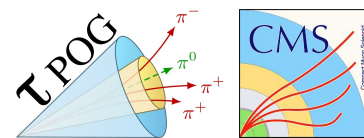
## Tight charge confidence



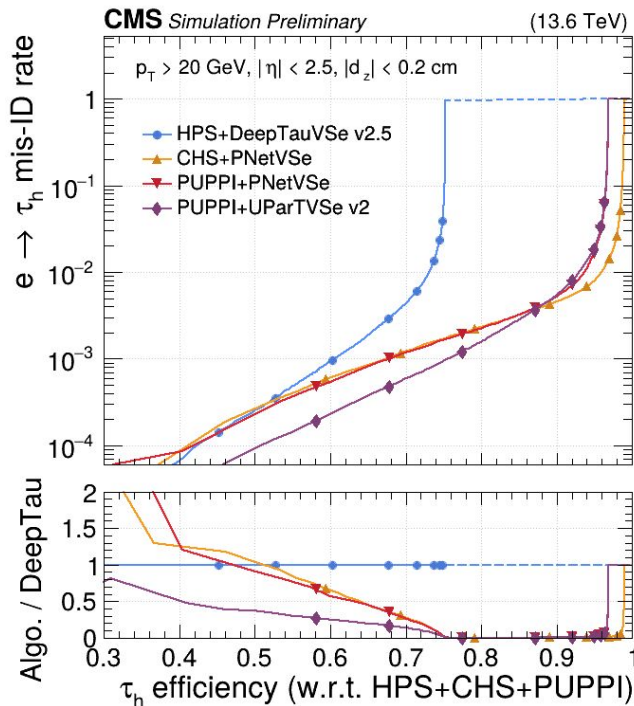
ROC curves illustrating performance of discrimination quark and gluon jets misidentified as  $\tau_h$  with HPS+DeepTau (blue, circle), CHS+PNet (orange, triangle up), PUPPI+PNet (red, triangle down), and PUPPI+UParT (purple, diamond) algorithms. In the left figure all  $\tau_h$  candidates are considered while in the right only those passing Tight charge confidence criteria.

The figures show that intrinsic efficiency of the HPS reconstruction is about 75% (70%) without (with) applying charge quality criteria (dashed blue line) and can be recovered with use of PNet/UParT. Globally the best performance is achieved by CHS+PNet followed by HPS+DeepTau for high purity regime used in many analyses.

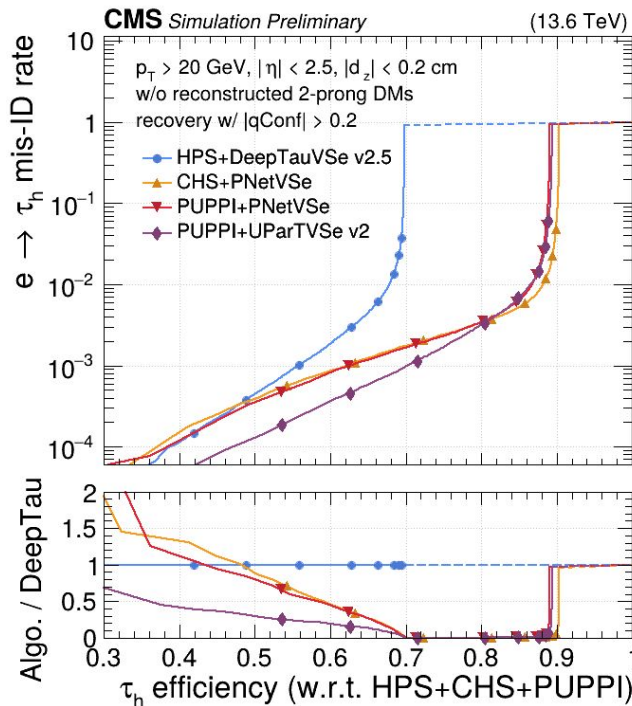
# Discrimination vs. $e \rightarrow \tau_h$ misID



## No charge confidence



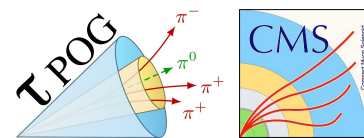
## Tight charge confidence



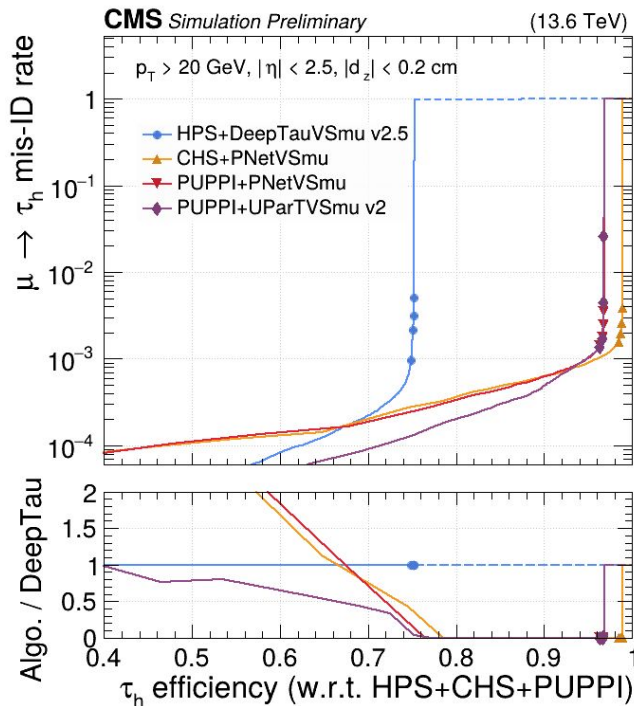
ROC curves illustrating performance of discrimination of electrons misidentified as  $\tau_h$  with HPS+DeepTau (blue, circle), CHS+PNet (orange, triangle up), PUPPI+PNet (red, triangle down), and PUPPI+UParT (purple, diamond) algorithms. In the left figure all  $\tau_h$  candidates are considered while in the right only those passing Tight charge confidence criteria.

The figures show that intrinsic efficiency of the HPS reconstruction is about 75% (70%) without (with) applying charge quality criteria (dashed blue line) and can be recovered with use of PNet/UParT. UParT achieves the best performance, followed by HPS+DeepTau in high-rejection mode, where its efficiency is not limited by the inefficiency of the HPS reconstruction.

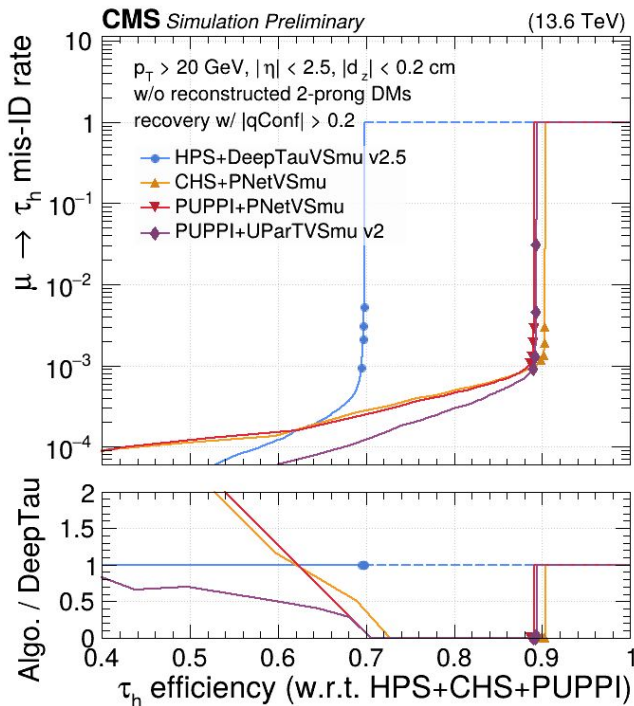
# Discrimination vs. $\mu \rightarrow \tau_h$ misID



## No charge confidence



## Tight charge confidence

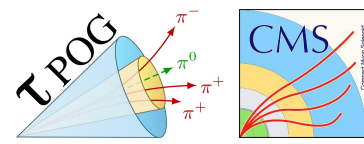


ROC curves illustrating performance of discrimination of muons misidentified as  $\tau_h$  with HPS+DeepTau (blue, circle), CHS+PNet (orange, triangle up), PUPPI+PNet (red, triangle down) and PUPPI+UParT (purple, diamond) algorithms. In the left figure all  $\tau_h$  candidates are considered while in the right only those passing Tight charge confidence criteria.

The figures show that efficiency of the HPS reconstruction is about 75% (70%) without (with) applying charge quality criteria (dashed blue line) and can be recovered with use of PNet/UParT.

UParT achieves the best performance, followed by HPS+DeepTau in high-rejection mode, where its efficiency is not limited by the inefficiency of the HPS reconstruction.

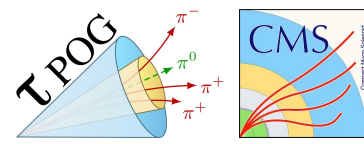
# Discriminant working points (VSjet)



To show performance representative for physics analyses, several working points (WPs) are defined independently for each discriminant (VSjet, VSe, or VSmu) targeting fixed efficiencies w.r.t. reconstruction of the  $\tau_h$  candidate with  $p_T > 20$  GeV,  $|\eta| < 2.5$ ,  $|d_z| < 0.2$  cm, and applying tight charge quality criteria. The reconstruction efficiency (including tight charge quality criteria) is defined w.r.t. the union of all HPS, CHS, and PUPPI  $\tau_h$  candidates with the same  $p_T$ ,  $\eta$ , and  $d_z$  requirements, but without any charge quality criteria. These WPs are shown in the respective ROC curves as solid markers. A sample of  $\tau_h$  candidates from an inclusive Drell–Yan sample with  $50 < m(\ell\ell) < 120$  GeV was used.

| Source           | Reconstruction |          | VSjet<br>discriminant | Rate w.r.t. reconstruction [%] |         |        |       |        |       |        |         |
|------------------|----------------|----------|-----------------------|--------------------------------|---------|--------|-------|--------|-------|--------|---------|
|                  | Algo.          | Eff. [%] |                       | VVVLoose                       | VVLoose | VLoose | Loose | Medium | Tight | VTight | VVTight |
| Genuine $\tau_h$ | HPS            | 70       | DeepTau v2.5          |                                |         |        |       |        |       |        |         |
|                  | CHS            | 90       | PNet                  | 98                             | 95      | 90     | 80    | 70     | 60    | 50     | 40      |
|                  | PUPPI          | 89       | PNet                  |                                |         |        |       |        |       |        |         |
|                  | PUPPI          | 89       | UParT v2              |                                |         |        |       |        |       |        |         |
| q/g jet<br>misID | HPS            | 15       | DeepTau v2.5          | 21                             | 13      | 7.9    | 4.2   | 2.5    | 1.5   | 0.92   | 0.53    |
|                  | CHS            | 32       | PNet                  | 29                             | 17      | 10     | 4.2   | 2.2    | 1.2   | 0.71   | 0.38    |
|                  | PUPPI          | 34       | PNet                  | 33                             | 19      | 12     | 5.6   | 3.0    | 1.7   | 0.95   | 0.49    |
|                  | PUPPI          | 48       | UParT v2              | 25                             | 15      | 8.9    | 4.6   | 2.6    | 1.5   | 0.82   | 0.47    |

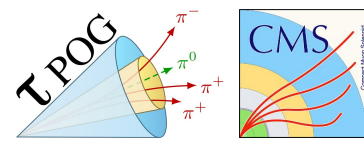
# Discriminant working points (VSe)



To show performance representative for physics analyses, several working points (WPs) are defined independently for each discriminant (VSjet, VSe, or VSmu) targeting fixed efficiencies w.r.t. reconstruction of the  $\tau_h$  candidate with  $p_T > 20$  GeV,  $|\eta| < 2.5$ ,  $|d_z| < 0.2$  cm, and applying tight charge quality criteria. The reconstruction efficiency (including tight charge quality criteria) is defined w.r.t. the union of all HPS, CHS, and PUPPI  $\tau_h$  candidates with the same  $p_T$ ,  $\eta$ , and  $d_z$  requirements, but without any charge quality criteria. These WPs are shown in the respective ROC curves as solid markers. A sample of  $\tau_h$  candidates from an inclusive Drell–Yan sample with  $50 < m(\ell\ell) < 120$  GeV was used.

| Source            | Reconstruction |          | VSe<br>discriminant | Rate w.r.t. reconstruction [%] |         |        |       |        |       |        |         |
|-------------------|----------------|----------|---------------------|--------------------------------|---------|--------|-------|--------|-------|--------|---------|
|                   | Algo.          | Eff. [%] |                     | VVVLoose                       | VVLoose | VLoose | Loose | Medium | Tight | VTight | VVTight |
| Genuine $\tau_h$  | HPS            | 70       | DeepTau v2.5        |                                |         |        |       |        |       |        |         |
|                   | CHS            | 90       | PNet                |                                |         |        |       |        |       |        |         |
|                   | PUPPI          | 89       | PNet                | 99.5                           | 99      | 98     | 95    | 90     | 80    | 70     | 60      |
|                   | PUPPI          | 89       | UParT v2            |                                |         |        |       |        |       |        |         |
| Electron<br>misID | HPS            | 91       | DeepTau v2.5        | 4.1                            | 2.5     | 1.5    | 0.67  | 0.33   | 0.11  | 0.041  | 0.016   |
|                   | CHS            | 93       | PNet                | 5.1                            | 2.4     | 1.3    | 0.63  | 0.40   | 0.22  | 0.12   | 0.061   |
|                   | PUPPI          | 94       | PNet                | 5.7                            | 2.7     | 1.4    | 0.65  | 0.37   | 0.19  | 0.10   | 0.050   |
|                   | PUPPI          | 94       | UParT v2            | 6.5                            | 3.1     | 1.6    | 0.72  | 0.36   | 0.12  | 0.045  | 0.020   |

# Discriminant working points (VSmu)

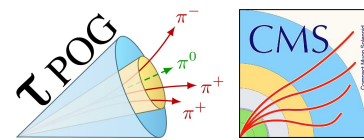


To show performance representative for physics analyses, several working points (WPs) are defined independently for each discriminant (VSjet, VSe, or VSmu) targeting fixed efficiencies w.r.t. reconstruction of the  $\tau_h$  candidate with  $p_T > 20$  GeV,  $|\eta| < 2.5$ ,  $|d_z| < 0.2$  cm, and applying tight charge quality criteria. The reconstruction efficiency (including tight charge quality criteria) is defined w.r.t. the union of all HPS, CHS, and PUPPI  $\tau_h$  candidates with the same  $p_T$ ,  $\eta$ , and  $d_z$  requirements, but without any charge quality criteria. These WPs are shown in the respective ROC curves as solid markers. A sample of  $\tau_h$  candidates from an inclusive Drell–Yan sample with  $50 < m(\ell\ell) < 120$  GeV was used.

| Source           | Reconstruction |          | VSmu<br>discriminant | Rate w.r.t. reconstruction [%] |       |        |       |
|------------------|----------------|----------|----------------------|--------------------------------|-------|--------|-------|
|                  | Algo.          | Eff. [%] |                      | VLoose                         | Loose | Medium | Tight |
| Genuine $\tau_h$ | HPS            | 70       | DeepTau v2.5         |                                |       |        |       |
|                  | CHS            | 90       | PNet                 | 99.95                          | 99.9  | 99.8   | 99.5  |
|                  | PUPPI          | 89       | PNet                 |                                |       |        |       |
|                  | PUPPI          | 89       | UParT v2             |                                |       |        |       |
| Muon<br>misID    | HPS            | 99       | DeepTau v2.5         | 0.53                           | 0.31  | 0.21   | 0.094 |
|                  | CHS            | 99       | PNet                 | 0.30                           | 0.19  | 0.13   | 0.12  |
|                  | PUPPI          | 99       | PNet                 | 0.29                           | 0.19  | 0.13   | 0.11  |
|                  | PUPPI          | 100      | UParT v2             | 3.1                            | 0.46  | 0.13   | 0.091 |

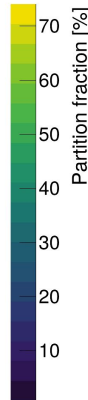


# Overlap & complement between $\tau_h$ ID (1/2)



HPS+DeepTau v2.5  
(75.0%)

CHS+PNet  
(96.9%)

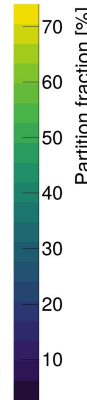


Genuine  $\tau_h$   
VVVLoose VSjet

PUPPI+UParT v2  
(95.9%)

HPS+DeepTau v2.5  
(68.5%)

CHS+PNet  
(88.4%)

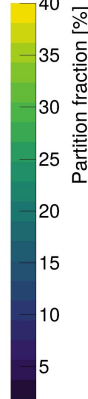


Genuine  $\tau_h$   
Medium VSjet

PUPPI+UParT v2  
(87.2%)

HPS+DeepTau v2.5  
(24.3%)

CHS+PNet  
(63.1%)

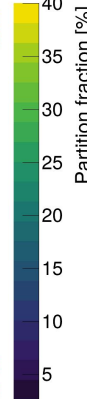


q/g  $\rightarrow$   $\tau_h$  mis-ID  
VVVLoose VSjet

PUPPI+UParT v2  
(81.2%)

HPS+DeepTau v2.5  
(30.5%)

CHS+PNet  
(55.7%)



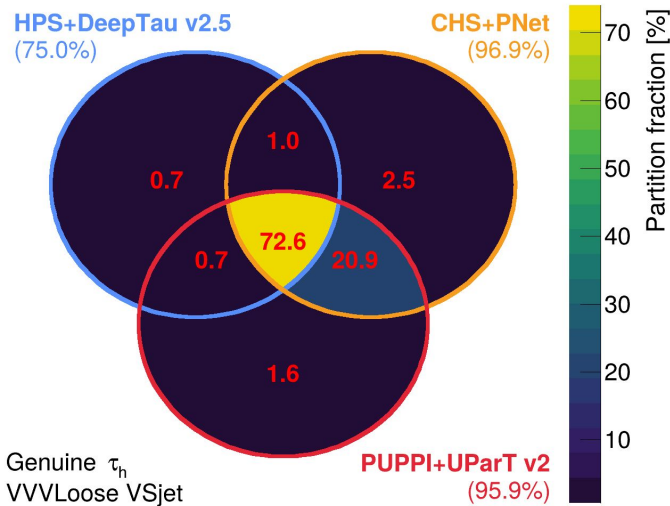
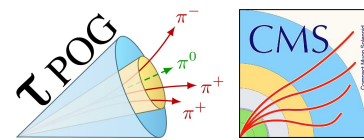
q/g  $\rightarrow$   $\tau_h$  mis-ID  
Medium VSjet

PUPPI+UParT v2  
(72.4%)

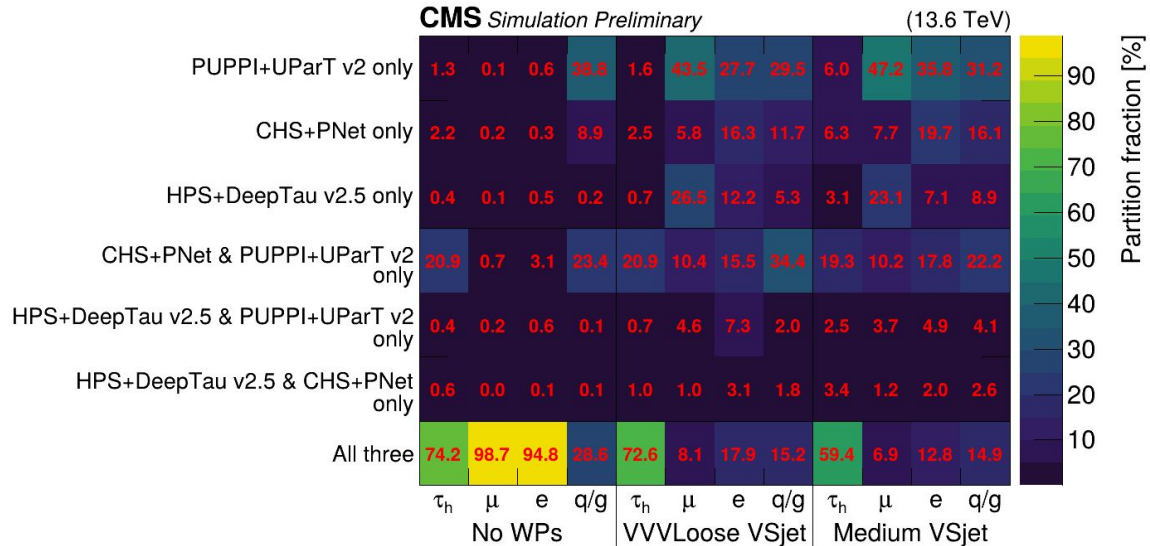
Venn diagrams illustrating the overlap and complement between  $\tau_h$  candidates from the HPS+DeepTau (blue circle), CHS+PNet (orange circle), PUPPI+UParT (red circle) algorithms. The red number and fill color of each partition indicate the fraction of candidates it contains relative to the union of all partitions in the diagram (i.e., the efficiency relative to the union). The Venn diagrams on the left (right) have the VVVLoose (Medium) VSjet WP applied, while all have the VVVLoose VSe and Loose VSmu WPs applied (see previous page). The top diagrams show genuine  $\tau_h$  candidates, while the bottom ones show jets misIDs. CHS+PNet and PUPPI+UParT can recover up to about 24–32% of genuine  $\tau_h$  candidates not reconstructed by the HPS algorithm or identified by DeepTau, but the rate of jet misIDs is higher.



# Overlap & complement between $\tau_h$ ID (2/2)

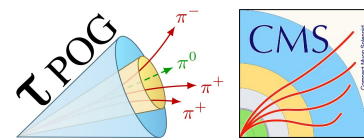


Venn diagram for genuine  $\tau_h$  candidates with the VVLoose VSjet WP applied (see previous page).



A table of the fractions in Venn diagram partitions versus genuine  $\tau_h$  or misidentified muons, electrons, or  $q/g$  jets. Different groups of VSjet WPs are shown in three columns: no WP, VVVLoose, or Medium. The VVVLoose VSe and Loose VSmu are applied when the VSjet is applied. While PNet and UParT can recover about 24–32% of genuine  $\tau_h$ , the rate of misidentified objects is relatively higher among the recovered  $\tau_h$  candidates (i.e., not reconstructed by HPS, and/or identified by DeepTau).

# Decay mode assignment



Reconstructed decay mode with HPS

**CMS Simulation Preliminary (13.6 TeV)**

| Reconstructed decay mode with HPS | None | $h^\pm$ | $h^\pm\pi^0$ | $h^\pm\geq 2\pi^0s$ | $h^\pm h^\mp h^\pm$ | $h^\pm h^\mp h^\pm\geq 1\pi^0s$ | Other |
|-----------------------------------|------|---------|--------------|---------------------|---------------------|---------------------------------|-------|
| Generated decay mode              |      |         |              |                     |                     |                                 |       |
| None                              | 0.12 | 0.29    | 0.36         | 0.15                | 0.25                | 0.44                            |       |
| $h^\pm h^\mp h^\pm (\pi^0s)$      | 0.00 | 0.01    | 0.02         | 0.17                | 0.11                | 0.03                            |       |
| $h^\pm h^\mp h^\pm \geq 1\pi^0s$  | 0.00 | 0.01    | 0.02         | 0.04                | 0.33                | 0.11                            |       |
| $h^\pm h^\mp h^\pm$               | 0.00 | 0.01    | 0.01         | 0.61                | 0.26                | 0.06                            |       |
| $h^\pm \geq 2\pi^0s$              | 0.00 | 0.00    | 0.00         | 0.00                | 0.00                | 0.00                            |       |
| $h^\pm \pi^0$                     | 0.09 | 0.53    | 0.54         | 0.01                | 0.04                | 0.33                            |       |
| $h^\pm$                           | 0.78 | 0.15    | 0.06         | 0.03                | 0.01                | 0.03                            |       |

Reconstructed decay mode with PNet

**CMS Simulation Preliminary (13.6 TeV)**

| Reconstructed decay mode with PNet | None | $h^\pm$ | $h^\pm\pi^0$ | $h^\pm\geq 2\pi^0s$ | $h^\pm h^\mp h^\pm$ | $h^\pm h^\mp h^\pm\geq 1\pi^0s$ | Other |
|------------------------------------|------|---------|--------------|---------------------|---------------------|---------------------------------|-------|
| Generated decay mode               |      |         |              |                     |                     |                                 |       |
| None                               | 0.09 | 0.15    | 0.19         | 0.11                | 0.15                | 0.22                            |       |
| $h^\pm h^\mp h^\pm (\pi^0s)$       | 0.00 | 0.00    | 0.00         | 0.00                | 0.00                | 0.00                            |       |
| $h^\pm h^\mp h^\pm \geq 1\pi^0s$   | 0.00 | 0.00    | 0.00         | 0.04                | 0.56                | 0.13                            |       |
| $h^\pm h^\mp h^\pm$                | 0.00 | 0.00    | 0.00         | 0.83                | 0.22                | 0.03                            |       |
| $h^\pm \geq 2\pi^0s$               | 0.00 | 0.04    | 0.34         | 0.00                | 0.01                | 0.22                            |       |
| $h^\pm \pi^0$                      | 0.09 | 0.72    | 0.45         | 0.01                | 0.05                | 0.30                            |       |
| $h^\pm$                            | 0.82 | 0.09    | 0.02         | 0.01                | 0.00                | 0.09                            |       |

Reconstructed decay mode with UParT

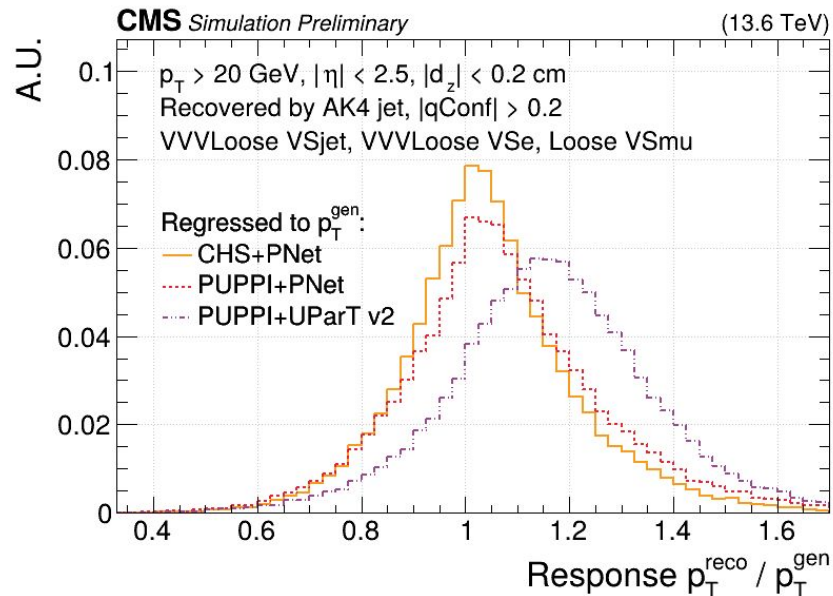
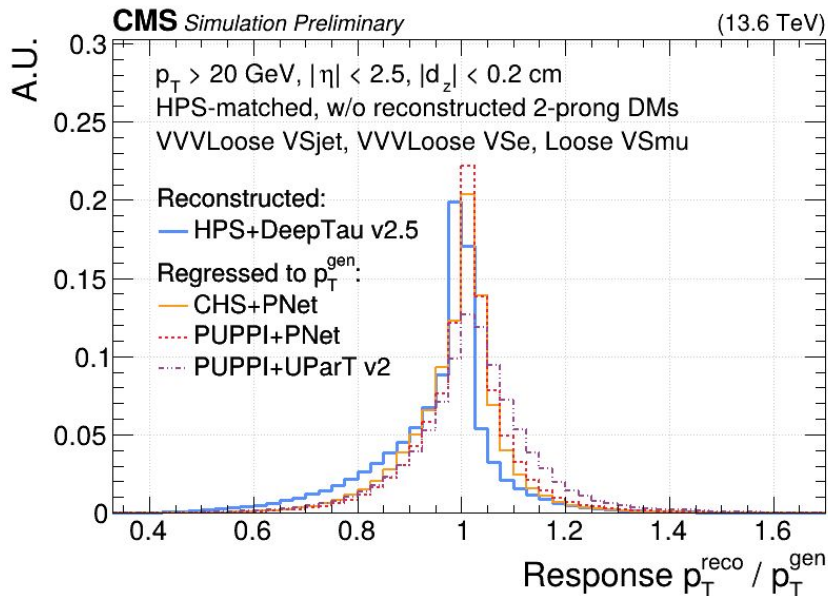
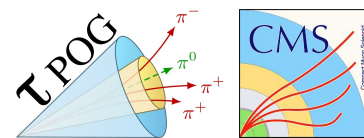
**CMS Simulation Preliminary (13.6 TeV)**

| Reconstructed decay mode with UParT | None | $h^\pm$ | $h^\pm\pi^0$ | $h^\pm\geq 2\pi^0s$ | $h^\pm h^\mp h^\pm$ | $h^\pm h^\mp h^\pm\geq 1\pi^0s$ | Other |
|-------------------------------------|------|---------|--------------|---------------------|---------------------|---------------------------------|-------|
| Generated decay mode                |      |         |              |                     |                     |                                 |       |
| None                                | 0.10 | 0.24    | 0.32         | 0.12                | 0.19                | 0.33                            |       |
| $h^\pm h^\mp h^\pm (\pi^0s)$        | 0.00 | 0.00    | 0.00         | 0.00                | 0.00                | 0.00                            |       |
| $h^\pm h^\mp h^\pm \geq 1\pi^0s$    | 0.00 | 0.00    | 0.00         | 0.07                | 0.63                | 0.13                            |       |
| $h^\pm h^\mp h^\pm$                 | 0.00 | 0.00    | 0.00         | 0.78                | 0.13                | 0.01                            |       |
| $h^\pm \geq 2\pi^0s$                | 0.00 | 0.07    | 0.33         | 0.00                | 0.01                | 0.21                            |       |
| $h^\pm \pi^0$                       | 0.15 | 0.63    | 0.34         | 0.02                | 0.04                | 0.25                            |       |
| $h^\pm$                             | 0.74 | 0.06    | 0.01         | 0.01                | 0.00                | 0.07                            |       |

Generator-level genuine  $\tau_h$  ( $p_T^{\text{gen}} > 20$ ,  $|\eta^{\text{gen}}| < 2.5$ ) matched to reconstructed  $\tau_h$  candidates

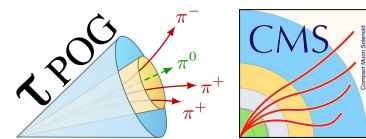
- PNet and UParT recover  $\tau_h$  lost by HPS
- PNet and UParT distinguish (with some precision) between  $h^\pm\pi^0$  and  $h^\pm\geq 2\pi^0s$  not resolved by HPS
- PNet and UParT have better resolution to distinguish between  $h^\pm h^\mp h^\pm$  and  $h^\pm h^\mp h^\pm\pi^0$  than HPS, and correctly assign decay mode for candidates with  $h^\pm h^\mp h^\pm(+\pi^0s)$  with a lost track, labeled as  $h^\pm h^\mp(+\pi^0s)$  for HPS

# The $p_T$ response



Distributions of the visible  $p_T$  response for generator  $\tau_h$  decay ( $p_T^{\text{gen}} > 20$  GeV and  $|\eta^{\text{gen}}| < 2.5$ ) reconstructed by HPS (blue solid), CHS (orange solid), or PUPPI (red dashed, or purple dash-dotted) and identified as a  $\tau_h$  by DeepTau, PNet, or UParT for the VVVLoose VSjet WP (98% efficiency), resp. The “reconstructed”  $p_T$  for PNet and UParT ( $p_T^{\text{reco}}$ ) is regressed to the generated visible  $p_T$ . **Left:** HPS objects, and CHS or PUPPI jets matched to a HPS candidate. **Right:**  $\tau_h$  candidates recovered as CHS or PUPPI jets by PNet or UParT (i.e., no HPS match).

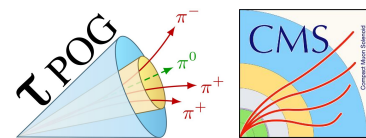
# Summary



This note reviews the machine learning (ML)-based classifiers used to identify genuine  $\tau_h$  against quark and gluon jets, electrons, and muons. Three  $\tau_h$  identification algorithms deployed by the CMS Collaboration during LHC Run 3 were compared: DeepTau v2.5, PNet and UParT v2. The note presents the performance of these algorithms in terms of genuine  $\tau_h$  efficiency versus the rate of  $\tau_h$  candidates from misidentified quark and gluon jets, electrons and muons; the accuracy of classifying  $\tau_h$  decay modes; and the  $p_T$  response.

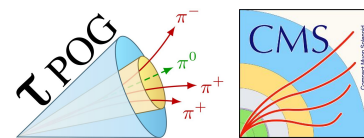
The different strengths of the tested classifiers — PNet provides the best rejection of misidentified quark and gluon jets, while DeepTau v2.5 and UParT v2 provide the best rejection of misidentified muons and electrons — mean that combining them yields the best tau identification to date.

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